CSCE 411
Linear Programming Practice

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1. You are in charge of a factory making robots out of recycled parts. Each robot requires k different types of components to assemble. In particular, it requires c_i copies of component i for each $1 \leq i \leq k$. There are ℓ different kinds of machines that you can acquire and recycle to get these components. For each $1 \leq i \leq \ell$, each unit of machine i costs d_i dollars and recycling it yields r_{ij} units of component j for each $1 \leq j \leq k$. You have a daily budget of b dollars to spend on buying machines to recycle (you may save money and components between days, however), and you want to maximize the average daily number of robots produced by your factory. Give a polynomial time algorithm to find the maximum possible average daily robot production.

2. Convert the LP below to equational form (do not attempt to solve the LP):

$$\begin{aligned} \text{minimize} \quad & x_1 + 2x_2 + 3x_3 \\ \text{subject to} \quad & 5x_1 + 3x_2 \leq 7 \\ & 6x_2 + 7x_3 \geq 4 \\ & x_1 \geq 9 \end{aligned}$$

3. Find the dual of the LP below. If the dual has a feasible solution with an objective value of 50, what can we say about the optimum objective value in the primal? (You do not need to solve the LP)

$$\begin{aligned} \text{maximize} \quad & x_1 + 2x_2 + 3x_4 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 \leq 7 \\ & x_1 + 3x_2 + x_4 \leq 14 \\ & 5x_2 + 4x_3 - 2x_4 \leq 26 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$

4. Solve this LP with the simplex method, using the pivot rule "pivot to whichever adjacent vertex increases the objective value the most" and starting from the basic feasible solution $x = (0, 3, 0, 1)$

$$\begin{aligned} \text{maximize} \quad & x_1 - x_2 + 3x_3 \\ \text{subject to} \quad & x_1 + x_2 + x_3 + x_4 = 4 \\ & x_1 + 2x_2 + 3x_3 + 4x_4 = 10 \\ & x_1, x_2, x_3, x_4 \geq 0 \end{aligned}$$